

To create two clusters of the eight patterns T1 to T8 given in Fig. 1.1, which have non-numeric attribute values, by the average-linkage procedure using Hamming distance, we begin by creating eight clusters {T1}, {T2}, {T3}, {T4}, {T5}, {T6}, {T7}, {T8}. The Hamming distances between the eight clusters is given by the following table:

Patterns	{T1}	{T2}	{T3}	{T4}	{T5}	{T6}	{T7}	{T8}
{T1}	0	2	1	3	1	2	2	2
{T2}	2	0	1	2	2	2	1	3
{T3}	1	1	0	2	2	1	1	3
{T4}	3	2	2	0	2	1	1	1
{T5}	1	2	2	2	0	3	1	1
{T6}	2	2	1	1	3	0	2	2
{T7}	2	1	1	1	1	2	0	2
{T8}	2	3	3	1	1	2	2	0

The smallest off-diagonal value in the preceding table is 1, and scanning the table row by row, we encounter this value for the first time between {T1} and {T3}. So, we merge these two clusters. The distances between the seven clusters we now have are shown in the following table:

Patterns	{T1, T3}	{T2}	{T4}	{T5}	{T6}	{T7}	{T8}
{T1, T3}	0	1.5	2.5	1.5	1.5	1.5	2.5
{T2}	1.5	0	2	2	2	1	3
{T4}	2.5	2	0	2	1	1	1
{T5}	1.5	2	2	0	3	1	1
{T6}	1.5	2	1	3	0	2	2
{T7}	1.5	1	1	1	2	0	2
{T8}	2.5	3	1	1	2	2	0

The smallest off-diagonal value in the preceding table is 1, and scanning the table row by row, we encounter this value for the first time between {T2} and {T7}. So, we merge these two clusters. The distances between the six clusters we now have are shown in the following table:

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Patterns	{T1, T3}	{T2, T7}	{T4}	{T5}	{T6}	{T8}
{T1, T3}	0	1.5	2.5	1.5	1.5	2.5
{T2, T7}	1.5	0	1.5	1.5	2	2.5
{T4}	2.5	1.5	0	2	1	1
{T5}	1.5	1.5	2	0	3	1
{T6}	1.5	2	1	3	0	2
{T8}	2.5	2.5	1	1	2	0

The smallest off-diagonal value in the preceding table is 1, and scanning the table row by row, we encounter this value for the first time between {T4} and {T6}. So, we merge the two clusters. The distances between the five clusters we now have are shown in the following table:

Patterns	{T1, T3}	{T2, T7}	{T4, T6}	{T5}	{T8}
{T1, T3}	0	1.5	2	1.5	2.5
{T2, T7}	1.5	0	1.75	1.5	2.5
{T4, T6}	2	1.75	0	2.5	1.5
{T5}	1.5	1.5	2.5	0	1
{T8}	2.5	2.5	1.5	1	0

The smallest off-diagonal value in the preceding table is 1, between {T5} and {T8}. So, we merge the two clusters. The distances between the four clusters we now have are shown in the following table:

Patterns	{T1, T3}	{T2, T7}	{T4, T6}	{T5, T8}
{T1, T3}	0	1.5	2	2
{T2, T7}	1.5	0	1.75	2
{T4, T6}	2	1.75	0	2
{T5, T8}	2	2	2	0

The smallest off-diagonal value in the preceding table is 1.5, between {T1, T3} and {T2, T7}. So, we merge the two clusters. The distances between the three clusters we now have are shown in the following table:

Patterns	{T1, T2, T3, T7}	{T4, T6}	{T5, T8}
{T1, T2, T3, T7}	0	1.9	2
{T4, T6}	1.9	0	2
{T5, T8}	2	2	0

The smallest off-diagonal value in the preceding table is 1.9, between {T1, T2, T3, T7} and {T4, T6}. So, we merge them to finally obtain two clusters: {T1, T2, T3, T4, T6, T7} and {T5, T8}.

FIGURE 10.9 Clustering of the eight professor-student patterns T1 to T8 given in Fig. 1.1 by the average-linkage procedure using Hamming distance.